

On the interpretation of hyperfine-structure intensity anomalies in the NH_3 (J, K)=(1, 1) inversion transition

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Summary. For NH_3 we present a simple radiative transfer model with selective trapping in the far IR (J, K)=(2, 1)→(1, 1) transition, which together with hyperfine selective collisional excitation is able to reproduce the observed hyperfine intensity anomalies in the molecules (J, K)=(1, 1) inversion transition. The model leads to the interpretation, that the anomalous hyperfine intensities emitting clouds are composed of many small (10^{-2} pc), high density (10^6 – 10^7 cm $^{-3}$) clumps of 0.3–1 $M_\odot$, which may be the precursors of low mass star formation.

Key words: molecular clouds – radiative transfer – star formation

1. Overview

Intensity anomalies in the nuclear quadrupole hyperfine structure of the NH_3 (J, K)=(1, 1) inversion transition, which were first detected in absorption toward the continuum source DR21 (Matsakis et al., 1977), are now known to be a rather common phenomenon in the NH_3 emission from medium warm interstellar clouds (Stutzki et al., 1982; Batrla et al., 1983; Morane, 1984; Stutzki et al., 1984). In the anomalous spectra the intensities of the outer satellites are not equal, as they would be in the case of equal excitation temperature in all hyperfine levels, but the $F_1=1\rightarrow0$ satellite is weaker and the $F_1=0\rightarrow1$ satellite is stronger than would be expected from the intensities of the inner satellites (the energy level scheme and the transitions involved are shown in Fig. 1). In some of the sources the inner satellites show very weak anomalous intensities with the $F_1=1\rightarrow2$ satellite having increased intensity and the $F_1=2\rightarrow1$ satellite being too weak.

Matsakis et al. (1977), proposed a mechanism producing the observed anomalies: the nonthermal population in the hyperfine states is induced by selective trapping in the hyperfine structure of the IR (2, 1)→(1, 1) transition. This requires the emitting gas to be clumped, with the individual clumps having small line widths (~ 0.3 – 0.6 km s $^{-1}$). We discuss this mechanism, which is the only one known capable of producing the observed anomalies, in Sect. 2 in more detail.

Based on this mechanism we did extensive radiative transfer calculations where we put special emphasis on keeping the model as simple as possible in order to have only a small number of adjustable parameters (T_k , n_{H_2} , N_{NH_3}). The radiative transfer model and the results of the numerical calculations are described in Sect.

3. As has been pointed out by Stutzki and Winnewisser (1984a) with nonthermal population in the hyperfine structure it is necessary to consider the exact hyperfine selective collision rates in the radiative transfer calculations.

With the results of the radiative transfer calculations we are then able to fit the observational data on anomalous NH_3 -spectra. The important result is, that it is possible to determine the H_2 -density in the emission region from the strength of the anomalies, whereas normal NH_3 -spectra only allow to determine the optical depth and rotational temperature, resp. column density and kinetic temperature (compare with Stutzki and Winnewisser, 1985). The result of these fits and their astrophysical implications are discussed in Sect. 4. The resulting interpretation is, that the anomalous NH_3 -emission comes from a number of small (diameter $\sim 10^{-2}$ pc) clumps with rather high densities (10^6 – 10^7 cm $^{-3}$), which have masses of 0.3–1.0 $M_\odot$. These clumps may perhaps be precursors of low mass stars.

A critical discussion of this result with special emphasis on the influence of the model assumptions and on possible independent observational tests is given in Sect. 5.

2. How the anomalies are produced

The mechanism producing the nonthermal excitation in the (J, K)=(1, 1) hyperfine structure levels was proposed by Matsakis et al. (1977) (and was also shown to give the same anomalies as the observed ones by means of radiative transfer calculations in the case of the absorption spectrum towards DR21). The central assumption is that the NH_3 -gas is clumped with the individual clumps having much smaller line widths (~ 0.3 km s $^{-1}$) than the observed width (~ 1 – 3 km s $^{-1}$). The observed line then is the average over several such clumps, where the emission from each clump shows a slightly different Doppler-shift due to the relative motion of the individual clumps.

Because of the small line width the outer hyperfine satellites ($F_1=1\rightarrow0$ and $F_1=2\rightarrow1$) of the IR (2, 1)→(1, 1) transition do not overlap with the inner satellites (see Fig. 1). The optical depth in these transitions is therefore lower than in the inner satellites, which overlap with the main line. Correspondingly, the escape probability β for photons in the outer satellites and therefore the effective radiative rates βA (A is the Einstein coefficient for spontaneous emission) are larger than those for the inner satellites, as these photons can no more be trapped in the inner satellites and the main line. This process is depicted schematically in Fig. 2. The result is that the final states of the outer hf-satellites, i.e. the $F_1=0$ and $F_1=1$ states, are somewhat overpopulated. Whereas the $F_1=0$ state of the upper inversion level can only decay via the F_1

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NH₃ Nuclear Quadrupole Hyperfine Structure

(J,K) = (1,1), (2,1)

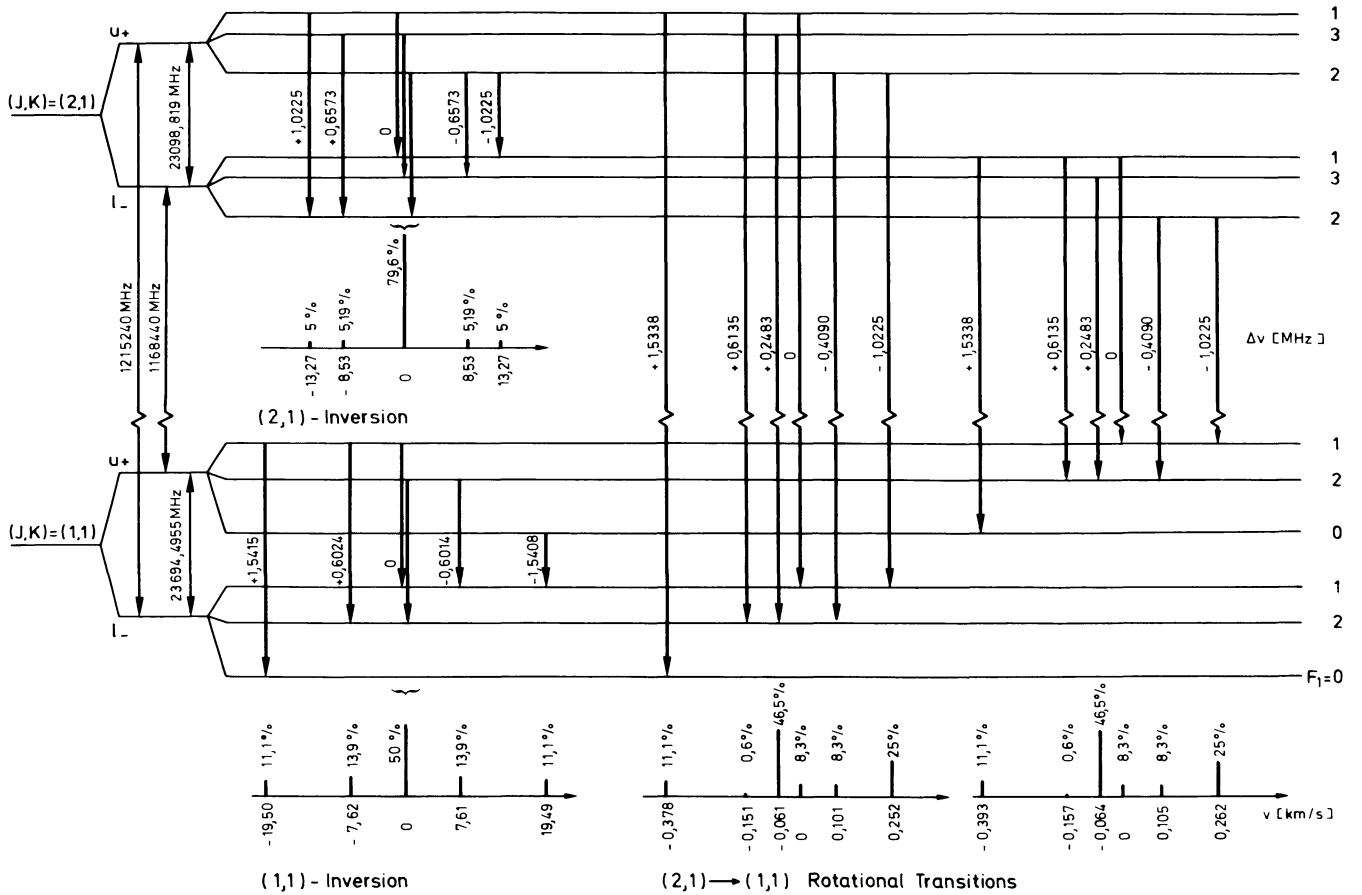


Fig. 1. Nuclear quadrupole hyperfine structure of the NH₃ (J, K) = (1, 1) and (2, 1) rotation inversion states. For each transition its position on the equivalent Doppler velocity scale and its natural intensity are given

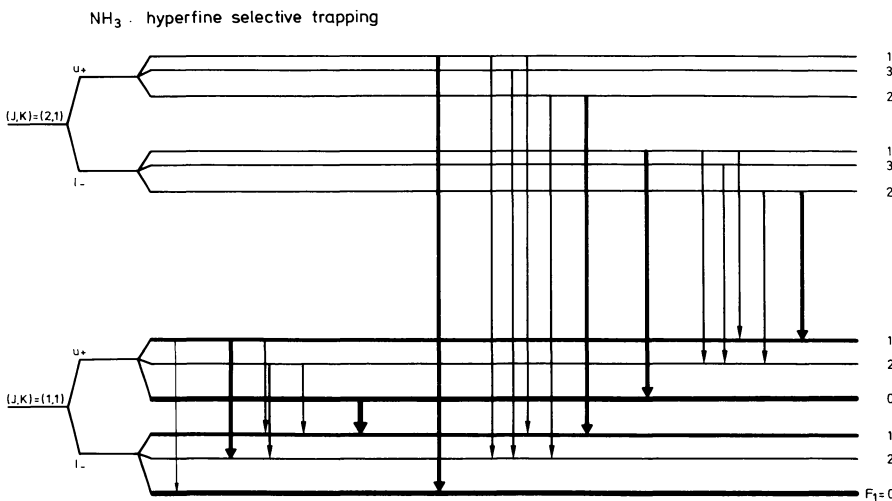


Fig. 2. The mechanism of hyperfine selective trapping inducing the observed anomalies is depicted schematically in this picture. The line intensities are indicated by the strength of the corresponding transition arrows, the nonthermal population of the (J, K) = (1, 1) hyperfine states is indicated by the strength of the level-lines

=0 → 1 transition into a state which is also slightly overpopulated, the $F_1 = 1$ state of the upper inversion level may decay via the $F_1 = 1 \rightarrow 2$, $1 \rightarrow 1$ and $1 \rightarrow 0$ transitions, i.e. especially into the normally populated $F_1 = 2$ state of the lower inversion level. Therefore the overpopulation is stronger for the $F_1 = 0$ levels than for the $F_1 = 1$ levels.

Considering the effect of this nonthermal population on the intensities of the satellites of the inversion transition, we see that the $F_1 = 0 \rightarrow 1$ satellite, going from a strongly overpopulated to a weaker populated level, is enhanced, whereas the $F_1 = 0 \rightarrow 1$ transition going to the strongly overpopulated $F_1 = 0$ level, is decreased in intensity. Similarly, the inner $F_1 = 1 \rightarrow 2$ satellite,

going from the (weakly) overpopulated $F_1=1$ to the normally populated $F_1=2$ state, is enhanced, and the $F_1=2 \rightarrow 1$ transition, going from the normally populated $F_1=2$ state to the (weakly) overpopulated $F_1=1$ state is decreased in intensity. In fact deviations of only several % from the thermal population are sufficient to produce the observed anomalies.

This line of arguments shows that selective trapping in the far IR rotation inversion $(J, K)=(2, 1) \rightarrow (1, 1)$ transition will lead to a qualitative explanation of the observed anomalies. Radiative transfer calculations, however, are necessary to give quantitative results. With nonthermal population being present in the hyperfine states, it is important to consider the hyperfine selective collisional excitation in detail as discussed by Stutzki and Winnewisser (1984). As the collisions always tend to thermalize the population, i.e. to reduce the anomalies, the strength of the resulting anomalies will depend on the H₂-density via the collision rates. The other factor influencing the strength of the anomalies is of course the linewidth of the individual clump, which determines the degree of overlap of the hyperfine satellites in the IR $(2, 1) \rightarrow (1, 1)$ transitions and therefore determines how strong the deviations from thermal population induced by selective trapping will be.

In principle similar anomalies as the ones observed in the $(J, K)=(1, 1)$ inversion transition should occur in other rotation-inversion levels. However for the next higher metastable level $(J, K)=(2, 2)$ the hfs-splitting in the corresponding IR $(3, 2) \rightarrow (2, 2)$ transition is much smaller ($\sim \pm 0.17 \text{ km s}^{-1}$) due to the accidental degeneracy of the $(J, K)=(3, 2)$ rotation-inversion state. For selective trapping effects to become important, therefore much smaller linewidths would be required. For still higher rotational levels (including non metastable states) due to the rapidly decreasing relative intensities of the $F \neq 0$ hyperfine satellites the hyperfine structure has never been observed, at least in sources, where the velocity structure would allow a clear identification of the satellites.

The thermal linewidth ($\Delta v_{\text{NH}_3} = 0.23 \text{ km s}^{-1} (T/20 \text{ K})^{1/2}$) is a lower limit on the width of the lines from interstellar molecules. With typical temperatures of 20–40 K the outer hyperfine satellites of the IR $(2, 1) \rightarrow (1, 1)$ transition are clearly separated from the inner satellites and the main line at the thermal line width, whereas the inner satellites fully overlap with the main component. The selective trapping effect therefore is most efficient at this line width. A broader line will produce weaker anomalies.

In order to reduce the number of adjustable parameters (see Sect. 4) and at the same time simplify somewhat the radiative transfer algorithm, we assumed a thermal line width of the individual clumps and treated the inner satellites and the main line of the IR $(2, 1) \rightarrow (1, 1)$ transition as a group of lines with equal frequency, whereas the outer satellites were treated as separate lines. We thus did our model calculations for the case of maximum anomalies; the influence of increasing the line width of the individual clumps on the results and the interpretation of the anomalous spectra will be discussed in Sect. 5.

3. The radiative transfer model

From the observations for each transition the intensity (given as main beam brightness temperature T_B), the line width Δv and LSR-velocity of the source are obtained. The latter parameter has no direct relation with the local physical properties of the emission region, but is determined by the large scale velocity structure of the cloud and its position in the galaxy. As described in the previous

section the observed line width must be interpreted as the overlap of the emission of the individual clumps with much smaller lines and therefore gives a measure for the average relative velocity of the clumps, but contains no information on the individual clumps. The ratio of the observed line width to the assumed line width for an individual clump gives a lower bound to the number of clumps K in the telescope beam

$$K \geq \frac{\Delta v_{\text{obs}}}{\Delta v}. \quad (1)$$

Because of the clumpiness of the emitting gas, the observed intensity is lower by the filling factor η_f than the intensity that would be observed for a clump filling the beam. Therefore out of the observational data only the relative intensities of the lines are related to the excitation conditions in the emitting region. For most of the anomalous sources this means, that only five observed quantities, the ratios of the intensity in the 4 satellites of the $(1, 1)$ -inversion line relative to the main component and the ratio of the $(2, 2)$ main line intensity relative to the $(1, 1)$ main line, can be used to determine the physical properties of the individual clumps (for some of the sources also the satellites in the $(2, 2)$ -inversion transition have been measured, which give four more observed parameters; their very small intensity is however known only with rather high observational errors, so that these extra parameters do not contribute any stringent additional information).

Thus it is necessary to keep the number of adjustable parameters in the radiative transfer model as small as possible to be able to determine the parameters by fits to the observed data. We choose the model of a homogeneous spherical cloud and treat the radiative transfer in an escape probability technique, which is described in more detail in Appendix A.

Of course the model of a homogeneous spherical cloud is not strictly self-consistent as the photon escape probability is larger near the surface of the cloud and smaller deep inside. Figure 3 shows the variation of the line escape probability [Eq. (A2)] with the distance from the clouds center for optical depths ranging from the optical thin to the optical thick case. With the escape probability in the cloud changing from place to place, also the

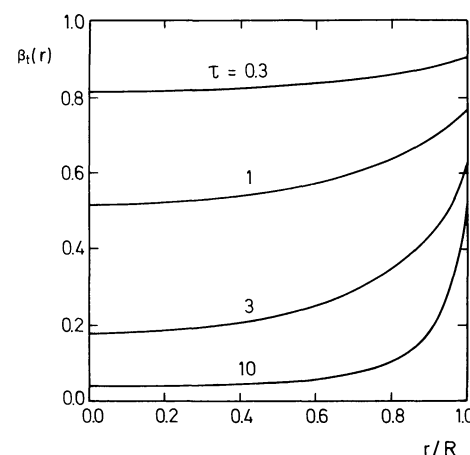


Fig. 3. The dependence of the line escape probability $\beta_l(r)$ [Eq. (A2)] on the position in a spherical symmetric homogeneous cloud is plotted for several optical depth values (the optical depth is calculated along the cloud-radius). For low optical depth $\beta_l(r)$ is only slightly less than unity in the whole cloud. For high optical depth it is very small, but nearly constant in the inner part of the cloud and increases rapidly to its limiting value of 0.5 at the cloud surface, where the photons can freely escape in one half-space

excitation conditions vary. In order to be consistent with the model of a homogeneous cloud one must choose some representative value of the escape probability to determine the radiative excitation processes for the whole cloud. What is commonly done and what we also choose to do, is to take the escape probability in the center of the cloud as this representative value:

$$\beta_t = \beta_t(r=0) = \int \frac{\phi_t(v-v_t)}{\Delta v_t} \exp(-\chi_v R) dv. \quad (2)$$

Here R is the radius of the cloud, t labels the transition, v_t is the center frequency of transition t , ϕ_t is the profile function, $\Delta v_t = \int \phi_t(v-v_t) dv$ the line width. The absorption coefficient is

$$\chi_v = \sum_t k_t \phi_t(v-v_t) \quad (3)$$

with

$$k_t = (n_{l_t} B_{l_t u_t} - n_{u_t} B_{u_t l_t}) \frac{h \nu_t}{4\pi} \frac{1}{\Delta v_t},$$

where u_t, l_t are the upper and lower level of transition t .

For the case of a Gaussian line profile (2) reduces to

$$\beta_t = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-z^2} e^{-\tau_G e^{-z^2}} dz, \quad (4)$$

where

$$\tau_G = \sum_g k_g R$$

is the total optical depth of all transitions g belonging to the same overlapping group G including transition t and measured along the radius R (if the transitions are not treated as having either the same frequency or not overlapping, then in addition to β_t we must consider the exchange probability for photons emitted in transition t to be absorbed in some overlapping transition t' [Eq. (A3); cf. Guilleoteau and Baudry (1981)]).

Approximating the escape probability in the cloud everywhere by its value at the center is not so bad an approximation. The reason is as follows: the observed inversion transitions and their hyperfine satellites show typical optical depths in the range 0.1–2. For this optical depth range the variation of the escape probability β with r is still rather slow (Fig. 3), so that the value at the center may well represent the region where most of the observed intensity originates. On the other hand, for the IR rotational transitions the optical depth is very high due to the low excitation of the upper states; but we are not interested in the outcoming intensity in these transitions, which due to the high optical depth will originate in the outer shell of the cloud with the excitation conditions being quite different from the center. Instead we are interested in their mean intensity deeper inside the cloud, which determines the excitation in the region where the observed inversion transition intensities originate. The mean intensity inside the cloud is however well described by the escape probability in the cloud center.

The mean intensity in one transition t belonging to the overlapping group of transitions G is given by (see Appendix A)

$$J_t = F_{v_t} \beta_t + S_G (1 - \beta_t), \quad (5)$$

where F_{v_t} is the background intensity at frequency ν_t , $S_G = \sum_g k_g S_g / \sum_g k_g$ is the source function of the transition group (g labels all transitions of that group) and the source function of one

transition is

$$S_g = \frac{2h\nu_g^3}{c^2} \frac{n_{u_g}/g_{u_g}}{n_{l_g}/g_{l_g} - n_{u_g}/g_{u_g}} \quad (6)$$

with the statistical weight g_i and the population n_i of the level i .

As the J_t depend themselves on the population via the source function and the escape probability, the system of rate equations

$$\begin{aligned} 0 = \frac{dn_i}{dt} = & \sum_{j \neq i} n_j n_{H_2} C_{ji} - \sum_{j \neq i} n_i n_{H_2} C_{ij} \\ & + \sum_t \{ \delta_{l_t, i} [(A_t + B_{u_t, i} J_t) n_{u_t} - B_{l_t, i} J_t n_{l_t}] \\ & - \delta_{u_t, i} [(A_t + B_{l_t, i} J_t) n_{l_t} - B_{u_t, i} J_t n_{l_t}] \}. \end{aligned} \quad (7)$$

is a set of nonlinear equations for the populations which may be solved by iterative solution of the linearized rate equations (for details of the linearization see Stutzki, 1984). Here B_{ij} are the Einstein coefficients for the induced radiative transitions, C_{ij} are the collision rate constants; the first line contains all collisional processes into and out of level i , the second one describes radiative processes going from a lower level to level i and the third line describes radiative transitions from an upper level to level i .

The emitting clumps (solid angle Ω_s) are much smaller than the beam (solid angle Ω_m), so that it is necessary to integrate the surface intensity of the cloud in order to calculate the observed intensity. The result for the main beam brightness temperature at the center of a line t is (see Appendix A)

$$T_{B,t} = \frac{c^2}{2k\nu_t^2} \frac{\Omega_s}{\Omega_m} (S_G - F_{v_t}) [1 - e(-2\tau_G)]; \quad (8)$$

$$\Omega_s \ll \Omega_m,$$

which differs from the commonly used detection equation

$$\begin{aligned} T_{B,t} = & \frac{c^2}{2k\nu_t^2} (S_G - F_{v_t}) (1 - \exp(-2\tau_G)); \\ \Omega_s \gg & \Omega_m, \end{aligned} \quad (9)$$

which gives the intensity coming out normal to the surface at the cloud center, by the presence of the filling factor Ω_s/Ω_m and by replacing the $\exp(-\tau)$ dependence with the function

$$e(\tau) = 2 \frac{1 - e^{-\tau(1+\tau)}}{\tau^2} \quad (10)$$

[that 2τ is written in (8) and (9) instead of τ results from the fact, that here the optical depth is calculated along the radius and not the diameter of the cloud]. Whereas the qualitative behaviour of $e(\tau)$ and $\exp(-\tau)$ is quite similar, the quantitative difference cannot be neglected when interpreting spectra with high S/N ratio as the observed anomalous spectra.

In the numerical calculations with this radiative transfer model we included all para NH_3 levels with $(J, K) = (1, 1), (2, 1), (3, 1), (2, 2), (3, 2)$, and $(4, 4)$, i.e. in total 36 levels including the inversion splitting and the hyperfine structure. The Einstein A -coefficients for the radiative transitions between these levels were calculated as described by Townes and Schawlow (1955), with a value for the dipole moment $1.476 \cdot 10^{-18}$ esu cm; the hyperfine selective collision rates were taken from Stutzki and Winnewisser (1985). The 77 radiative transitions between these levels (neglecting the very weak, forbidden $K \pm K' = \pm 3$ transitions) were grouped in 12

overlapping groups and 28 single lines according to the assumed line width of 0.3 km s^{-1} . Neglecting rotational levels higher than the $(J, K) = (4, 4)$ level does not restrict the validity of the results, as higher levels show only very low population even at the highest considered temperature of $T_k = 40 \text{ K}$. Comparison calculations without the $(4, 4)$ level show that even this has only very small influence on the lower levels (maximal deviations are of the order of 0.5% in the resulting intensities). Selective trapping effects in the IR $(3, 1) \rightarrow (2, 1)$ line are also of minor importance. Comparison calculations where the hyperfine structure of this transition was treated as one overlapping group of lines result in relative variations of the population in the $(2, 1)$ levels of 0.2% in the worst case. Thus we conclude, that the essential excitation processes occur in the $(1, 1)$, $(2, 1)$ and $(2, 2)$ transition.

Starting with thermal population in all levels, the iterative solution of the linearized rate equations usually takes 5–20 steps until the variations of the population are less than 0.1% in three subsequent steps (one iteration takes 0.1 s on a CYBER 176). We calculated a model grid of roughly 1000 parameter points for the parameter range $T_k = 18\text{--}40 \text{ K}$, $n_{\text{H}_2} = 10^{3.5}\text{--}10^7 \text{ cm}^{-3}$ and

$$N_{\text{NH}_3}/\Delta v = 10^{14.2}\text{--}10^{15.6} \text{ cm}^{-2} \text{ km s}^{-1}.$$

Figure 4 shows as the results of this calculations the observable relative intensities and the brightness temperature of the $(1, 1)$ main component, which is, due to the filling factor uncertainty, an upper bound on the observed intensity.

The anomalies in the outer satellites are strongest in the density range of $10^{4.5}\text{--}10^{5.5} \text{ cm}^{-3}$ for the $F_1 = 0 \rightarrow 1$ satellite, which is too strong, and in the range of $10^5\text{--}10^{6.5} \text{ cm}^{-3}$ for the $F_1 = 1 \rightarrow 0$ satellite, which is too weak. At very high column densities, corresponding to an optical depth > 10 , the intensity of the $F_1 = 0 \rightarrow 1$ satellite may even become greater than the main line intensity, a case which has not been observed until now. Another interesting effect is that at higher temperatures ($> 30 \text{ K}$) the relative intensity of the $F_1 = 1 \rightarrow 0$ satellite may drop below its thermal equilibrium minimum value of 0.22.

The anomalies in the inner satellites are much weaker and are not, in contrast to the outer satellites, enhanced in a special density range. They increase with increasing temperature and column density. The relative intensity of the $(2, 2)$ line and the intensity of the $(1, 1)$ transition show no significant deviations from the case without selective trapping (cf. Stutzki and Winnewisser, 1985).

An important question is, in how far the usual analysis applied to normal NH₃ spectra is applicable to spectra with anomalies. In the standard analysis the optical depth is derived under the assumption of equal T_{ex} in the hyperfine satellites from the relative intensity of the satellites, which according to (8) is

$$\frac{T_{B;s}}{T_{B;m}} = \frac{1 - e(2\tau_s)}{1 - e(2\tau_m)}. \quad (11)$$

With the presence of anomalies there is of course no chance to derive reasonable values of τ from the intensities of the outer satellites as is shown in Fig. 5. The weaker anomalies in the inner satellites, however, still allow for deriving the optical depth from the averaged intensity of both satellites (Fig. 6). From the intensity ratio of the $(2, 2)$ and the $(1, 1)$ main line and the optical depth of the $(1, 1)$ line it is possible to determine the rotational temperature between these levels with the same restrictions as in the standard analysis (see Stutzki and Winnewisser, 1984b). As there, however, it is not possible to derive the kinetic temperature from the rotational temperature without knowledge of the H₂-density.

4. Interpretation of the observed anomalous spectra

Whereas in the standard analysis of NH₃-spectra it is not possible to determine the H₂-density from relative intensities alone, i.e. without knowledge of the beam filling factor, in the case of anomalies the variation of the strength of the anomalies with the H₂-density allows for an independent determination of the H₂-density. As the observed spectra result from the integrated intensity of several clumps in the telescope beam, the clump properties derived in the following must be understood as average values for these clumps.

It is however a hopeless task to try to formulate the relation between the observed relative intensities and the three model parameters T_{kin} , n_{H_2} and N_{NH_3} in an analytic formula. Nevertheless, the parameters may be determined by least square fits to the observational data. The data have been summarized by Stutzki et al. (1984); here we give fit results of those spectra for which the observational accuracy allowed a determination of the parameters to better than 30%. The fit results together with the fit uncertainties (1σ -errors) are listed in Table 1a. Almost all observed spectra can be fitted within the observed errors ($\chi^2 < 1$), especially the spectra with very high signal/noise ratio as S106 ($\Delta\alpha, \Delta\delta$) = (0, 0), (160, 0), (0, 40), (0, -40), OMC S3, S4, OMC2, W48 (-40, 80). Only few of the fits have a χ^2 greater than 1, e.g. those at S106 (200, 40), (-40, 0), the 23.5 km s^{-1} component in S87 with only weak anomalies, OMC S1, S2 and W48 (-40, 80). The kinetic temperatures typically range from 20–25 K, the densities are rather high ($10^6\text{--}10^7 \text{ cm}^{-3}$) and the column densities are in the range $10^{14}\text{--}10^{15} \text{ cm}^{-2}/\Delta v [\text{km s}^{-1}]$. The very high density in the case of OMC S3 and S4 perhaps overestimate the actual density by one order of magnitude, as is described in Sect. 5.

Some of the spectra admit a second solution [comparable to the ambiguity which is present in the interpretation of normal NH₃-spectra, see Stutzki and Winnewisser (1985)]. These fits are listed in Table 1b. The typical density for these solutions is only $10^{4.5} \text{ cm}^{-3}$, i.e. two orders of magnitude lower than the other solution, the temperature is somewhat higher ranging from 25–30 K. Some of these second solutions are in contradiction to the observations, as the predicted brightness temperatures are lower than the observed ones [i.e. the formally defined filling factor $\eta_f = T_B(\text{observed})/T_B(\text{calculated})$ is greater than 1]. As these fits are only possible for some of the observed spectra and in addition have somewhat higher mean square deviations, it seems probable that the first class of solutions is the right one. That this is indeed the case will be shown below.

The collision rates used in the radiative transfer calculations are calculated for the case of NH₃-He-collisions. The NH₃-H₂-rates are estimated to be a factor $\alpha = 1.5$ higher than the NH₃-He-rates (Green, 1981). Thus the derived H₂-densities must be corrected accordingly. Let us denote in the following by n' the H₂-density derived from the radiative transfer calculation. We then have

$$C_{\text{He}} n' = C_{\text{He}} n_{\text{He}} + C_{\text{H}_2} n_{\text{H}_2}.$$

Assuming a helium abundance of 4 H₂-molecules per He-atom ($X_{\text{He/H}_2} = 0.25$) we then get

$$n_{\text{H}_2} = \frac{n'}{X_{\text{He/H}_2} + \alpha} = \frac{1}{1.75} n',$$

and for the mass density

$$\begin{aligned} \varrho &= m_{\text{H}_2} n_{\text{H}_2} + m_{\text{He}} n_{\text{He}} \\ &= (2 + 4X_{\text{He/H}_2}) m_{\text{H}} n_{\text{H}_2} = 1.71 m_{\text{H}} n'. \end{aligned}$$

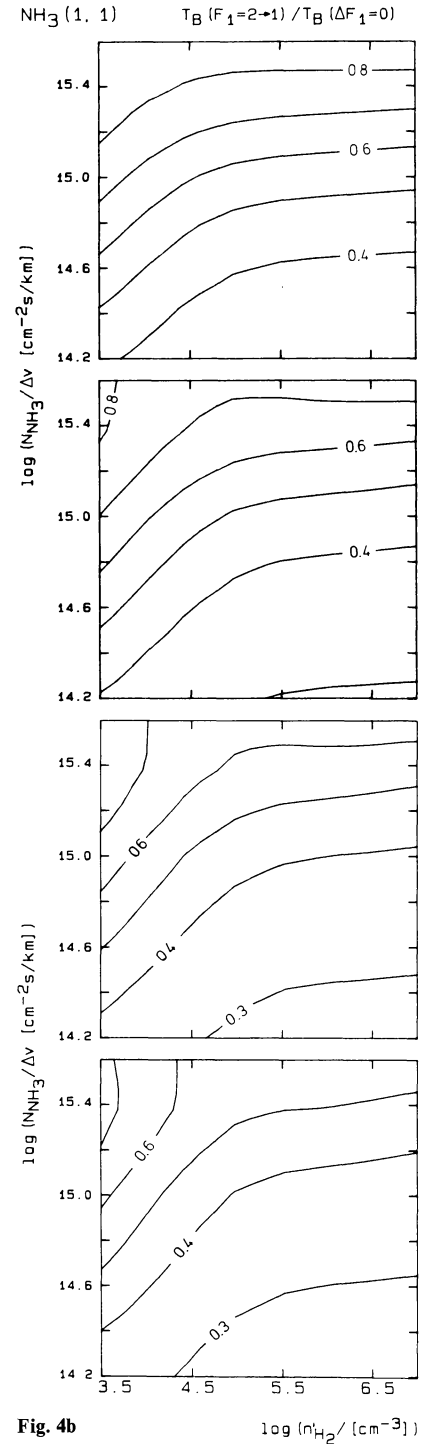
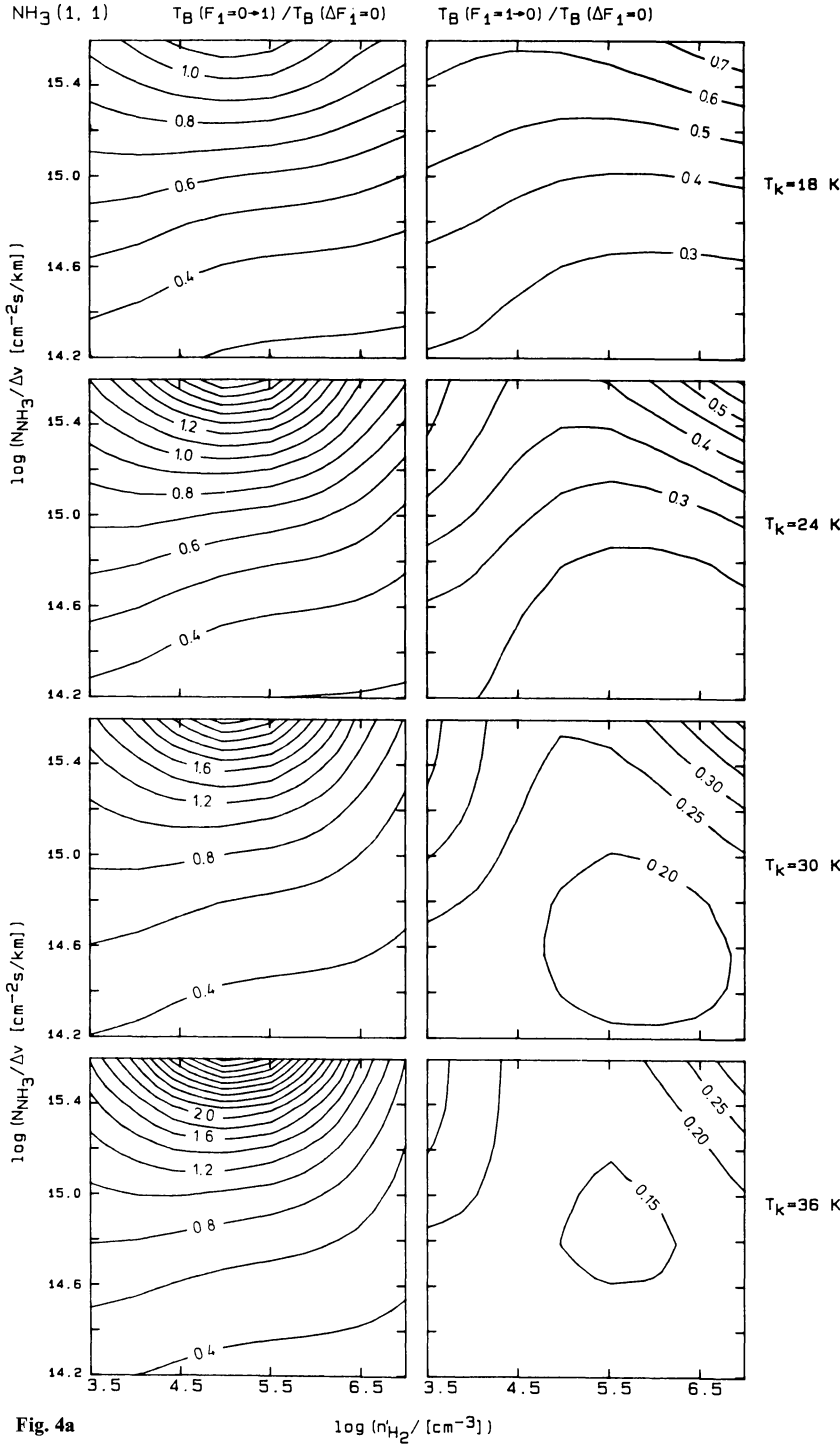


Fig. 4a

Fig. 4b

Fig. 4a–c. The results of the radiative transfer model calculations are shown as contour plots in the $\log(n_{\text{H}_2})/\log(N_{\text{NH}_3})$ plane for kinetic temperatures of 18, 24, 30, and 36 K (from top to bottom). **a** shows the relative intensity of the outer $F_1 = 0 \rightarrow 1$ (1, 1) satellite on the left and the $F_1 = 1 \rightarrow 0$ satellite on the right, **b** is the same for the inner (1, 1) satellites, and **c** gives the relative intensity of the (2, 2) main line on the left and the brightness temperature of the (1, 1) main line on the right

Knowing the density and temperature of the anomalously emitting clumps we can calculate the Jeans length and Jeans mass of one clump (Lang, 1980)

$$\lambda_J = \sqrt{\frac{\pi k T}{\mu m_{\text{H}} G \rho}} = 0.776 \cdot 10^{-3} \text{ pc} \left(\frac{T_1}{n'_7} \right)^{1/2}$$

$$M_J = \frac{\pi}{6} \rho \lambda_J^3 = 0.103 M_{\odot} \left(\frac{T_1^3}{n'_7} \right)^{1/2}$$

(T_1 is the kinetic temperature in units of 10^4 K, n'_7 is the density in units of 10^7 cm^{-3}). The derived Jeans masses for the high density fits (Table 2a) are $0.3\text{--}1 M_{\odot}$, for the low density fits (Table 2b) $\sim 10 M_{\odot}$, the Jeans lengths are 10^{-2} pc resp. $\sim 1 \text{ pc}$.

Knowing the model parameters from the fit, we also know the brightness temperature an individual clump should have. With the filling factor $\eta_c = \Omega_c/\Omega_m$ of one individual clump we then get for the observable frequency integrated intensity $K T_B \Delta \nu \eta_c$, where K is the

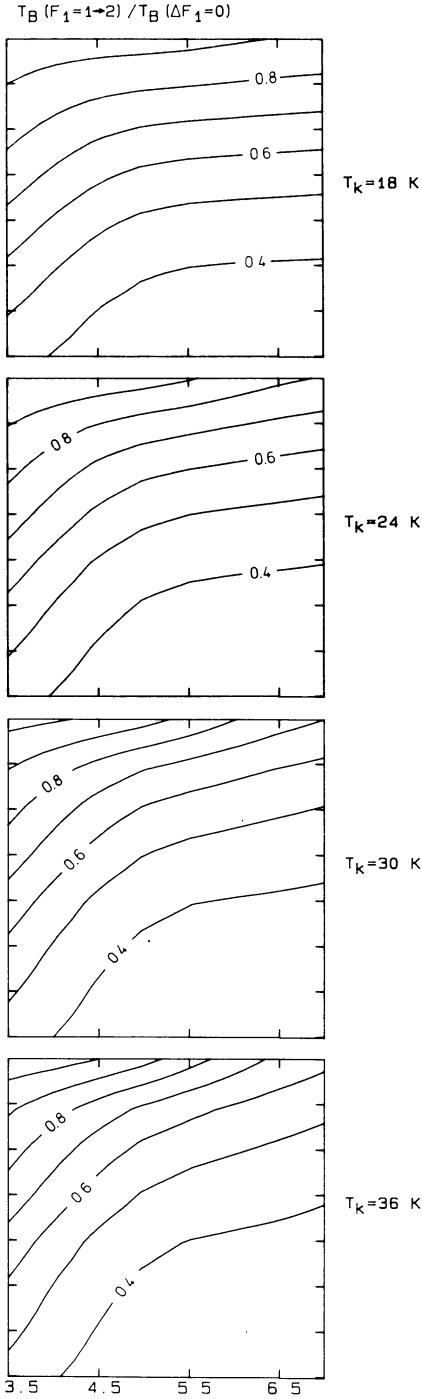


Fig. 4b (continued)

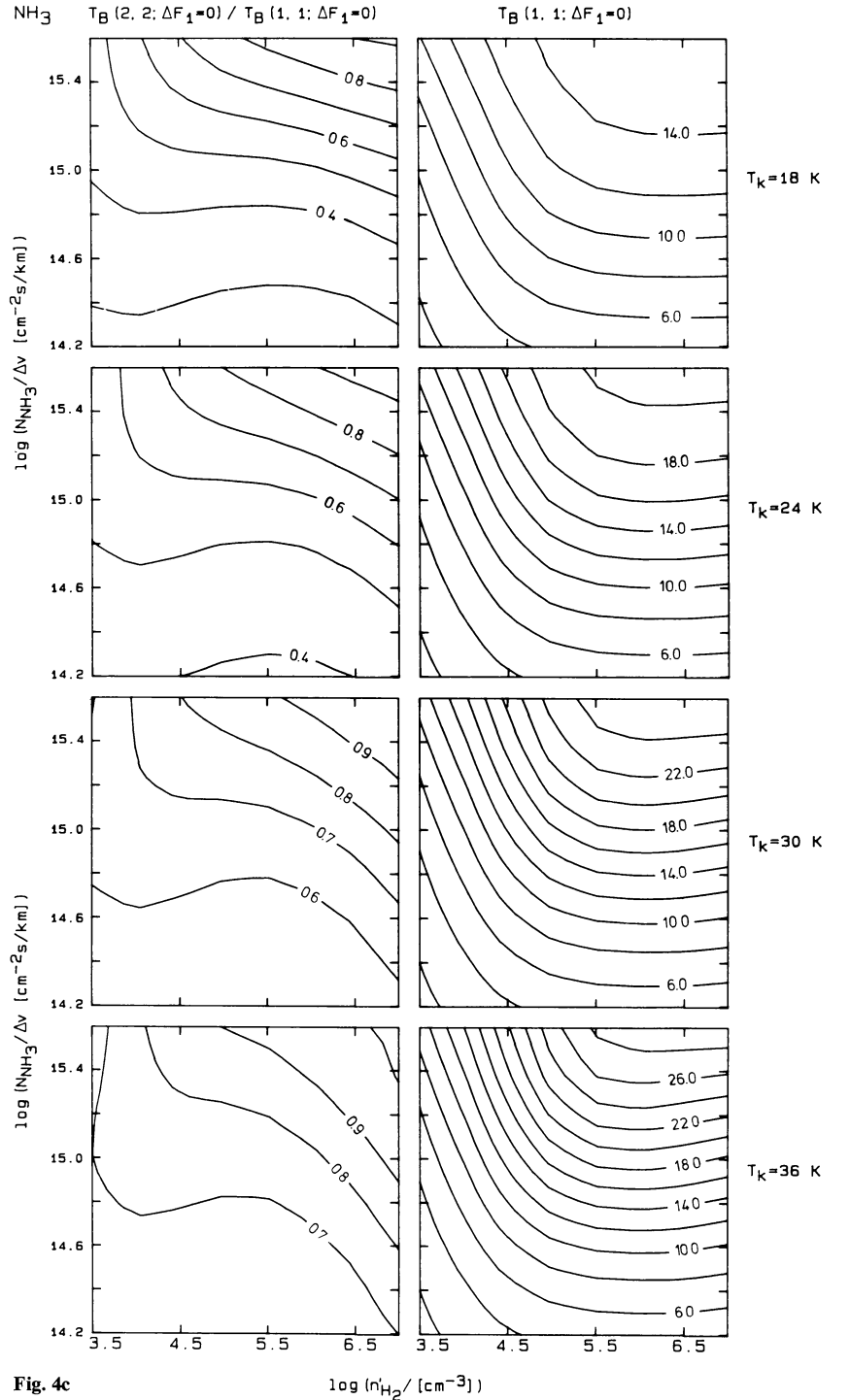


Fig. 4c

log($n'_{\text{H}_2}/\text{cm}^{-3}$)

number of clumps in the telescope beam [see Eq. (1)]. Comparing this with the actually observed integrated intensity $T_{B_{\text{obs}}}\Delta v_{\text{obs}}$ we get

$$\eta_c = \kappa \eta_f \leq \eta_f.$$

$\eta_f = T_{B_{\text{obs}}}/T_B$ is tabulated in Table 1. With the known distance r of the source and the beam size of the telescope $\Omega_m = \pi/(4 \ln 2) \theta_m^2$ ($\theta_m = 40''$ for the Effelsberg 100 m dish at NH₃-inversion frequencies) we are then able to calculate the radius of one clump from the

filling factor

$$R_c = \left(\frac{r^2 \Omega_m}{\pi \eta_f \kappa} \right)^{1/2},$$

and with the known density we get the mass of one clump

$$M_c = \frac{4\pi}{3} R_c^3 \rho = \kappa^{3/2} M_c(\text{max})$$

$$M_c(\text{max}) = 347.4 M_\odot n' r_{0.5 \text{ kpc}}^3 (\eta_f)^{3/2}.$$

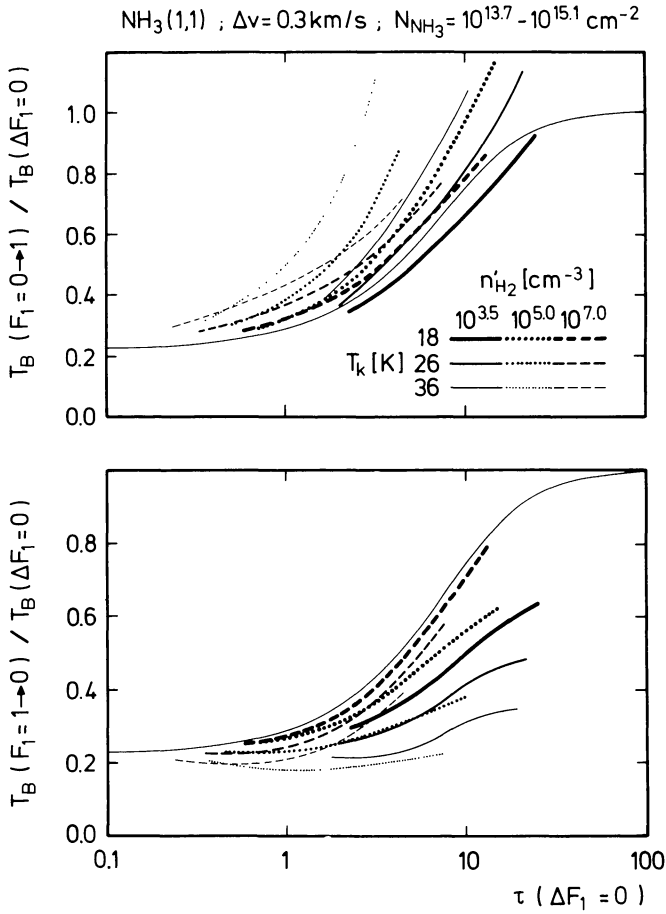


Fig. 5. The relative intensity of the outer satellites of the (1,1) inversion line is plotted against the optical depth of the main component for different values of the kinetic temperature and H_2 density. The thin solid line is the curve expected from Eq. (11). Along each curve shown, the NH_3 column density varies from $10^{13.7}$ to $10^{15.1} \text{ cm}^{-2}$.

For the high density fits $M_c(\text{max})$, which is the maximum mass of one individual clump (if $\kappa=1$), lies in the range $1\text{--}100 M_\odot$ for the sources located near to the sun, i.e. S106, S87, and OMC. For W48 due to the larger distance there is more emitting material in the beam, so that $M_c(\text{max}) \sim 10^3 M_\odot$. For the low density fits $M_c(\text{max})$ is $\sim 1 M_\odot$ resp. $10 M_\odot$.

Up to here the derived results are solely determined by the underlying model of a clumped cloud with the individual clumps having very narrow lines. To proceed further we must assume in addition, that the individual clumps have masses roughly equal to the Jeans mass. The small line width of the individual clumps (near the thermal line width) means, that no systematic or turbulent velocity fields are present, i.e. the clumps are nearly stable. The mass therefore cannot exceed the Jeans mass. On the other hand the presence of clumps with masses lower than the Jeans mass seems to be improbable, as no mechanism is known to produce such low mass fragments. Putting thus $M_c = M_J$ we are able to calculate κ and the number of clumps per beam K

$$K = \frac{1}{\kappa} \frac{\Delta v_{\text{obs}}}{\Delta v} = \left(\frac{M_c(\text{max})}{M_J} \right)^{2/3} \frac{\Delta v_{\text{obs}}}{\Delta v}$$

$$= 225 \frac{\eta_f n'_f}{T_l} r_{0.5 \text{ kpc}}^2 \frac{\Delta v_{\text{obs}}}{\Delta v}.$$

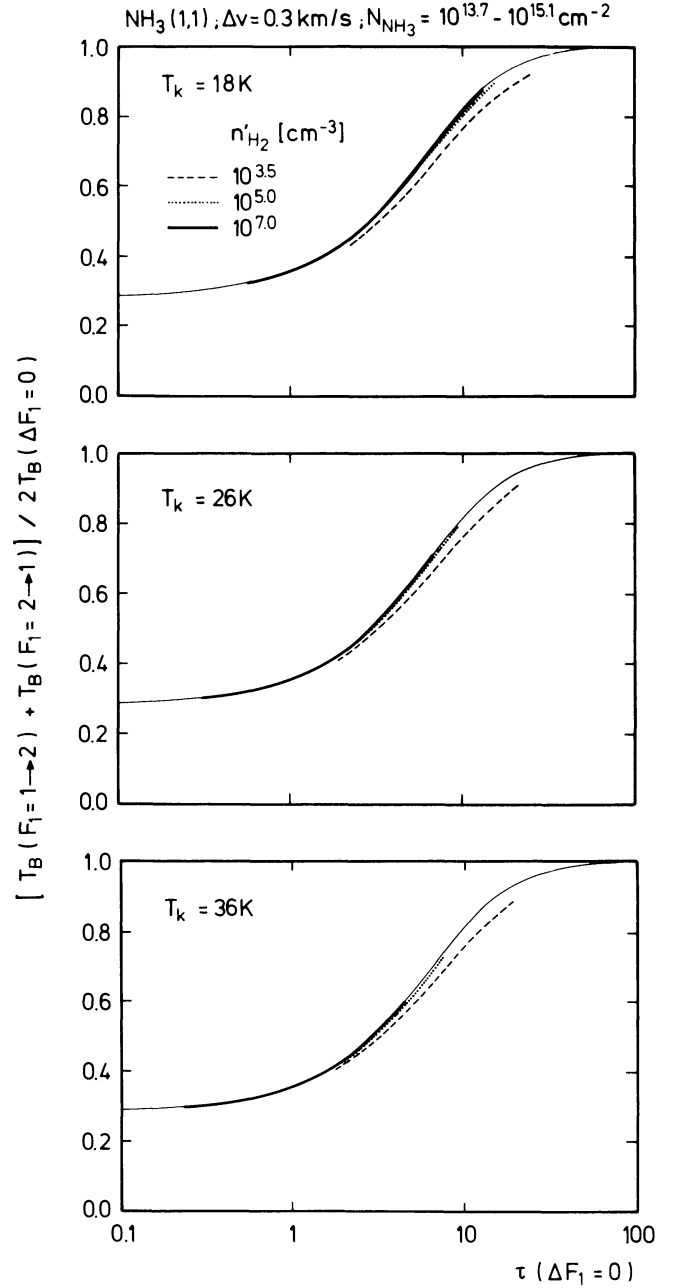


Fig. 6. Same as Fig. 5, but for the averaged relative intensity of the inner satellites of the (1,1) line. Their averaged relative intensity is quite well described by Eq. (11) (thin solid line)

For the high density fits there are some 10–100 clumps per beam for the nearer sources and up to 1000 clumps per beam in the case of W48 (Table 2a).

For the low density solutions the number of clumps calculated in this way is ~ 1 or even less, which is in contradiction to the observed linewidth requiring at least $K = \Delta v_{\text{obs}} / (0.3 \text{ km s}^{-1}) \sim 5$ clumps per beam. The low density solutions therefore are not selfconsistent. This confirms the suggestion that out of the two solutions only the high density solution is relevant.

From the number of clumps per telescope beam we can deduce the number density v of clumps by assuming in addition that the line of sight extension of the whole cloud corresponds to its total

Table 1a. Fit results to the observed spectra: $\Delta v = 0.3 \text{ km s}^{-1}$; high-density solution

Position	T_k (K)	$\log(n_{\text{H}_2}^-/\text{cm}^{-3})$	$\log(N_{\text{NH}_3}/\text{cm}^{-2})$	χ^2	Δv_{obs} (km/s)	T_B (K)		η_f
						theor.	obs.	
S106								
200, 40	21.89(0.82)	6.528(0.158)	14.29(0.026)	8.61	1.43(0.08)	12.705	3.04(0.06)	0.237
200, 80	17.72(3.42)	7.099(1.318)	14.33(0.106)	0.27	1.52(0.12)	11.044	1.32(0.09)	0.120
160, 40	23.78(1.71)	6.770(0.315)	14.23(0.058)	0.92	1.91(0.07)	12.045	2.49(0.08)	0.207
160, 0	23.05(1.11)	6.296(0.360)	14.12(0.040)	0.29	1.58(0.04)	10.487	1.96(0.04)	0.187
90, 30	34.25(4.09)	4.960(1.482)	13.78(0.286)	0.32	2.37(0.14)	5.484	0.55(0.03)	0.100
40, 0	27.46(2.59)	6.499(0.416)	14.23(0.097)	0.70	1.58(0.07)	12.703	0.84(0.03)	0.066
0, 40	21.91(0.85)	6.537(0.169)	14.26(0.028)	0.46	1.81(0.05)	12.255	1.38(0.03)	0.113
0, 0 (a)	21.81(0.57)	6.255(0.167)	14.36(0.019)	0.88	1.47(0.02)	13.708	4.71(0.06)	0.344
0, -40	22.54(5.80)	4.839(3.430)	14.07(0.717)	0.28	2.00(0.19)	8.150	1.42(0.12)	0.174
-40, 0	20.73(0.50)	5.496(0.481)	14.31(0.029)	4.83	1.21(0.02)	12.286	3.57(0.04)	0.291
S87								
0,0 21.0 km/s	23.99(1.85)	6.477(0.243)	14.47(0.046)	0.57	1.36(0.07)	15.936	1.10(0.04)	0.069
0,0 23.5 km/s	26.03(1.57)	7.258(0.211)	14.30(0.052)	4.23	2.05(0.07)	12.968	1.28(0.03)	0.099
W48								
-80, 40	20.76(1.91)	6.143(0.983)	14.16(0.067)	1.08	1.59(0.07)	10.567	1.56(0.06)	0.148
-80, 80	18.88(1.07)	7.227(0.473)	14.07(0.040)	0.50	1.72(0.04)	8.827	2.60(0.05)	0.295
-40, 40	22.34(1.31)	6.830(0.220)	14.37(0.038)	1.30	2.66(0.09)	13.696	1.40(0.04)	0.102
0, 40 45km/s	32.20(3.83)	5.525(1.477)	14.41(0.181)	1.59	1.95(0.16)	16.768	1.01(0.06)	0.060
0, 80	25.68(2.28)	6.425(0.353)	14.42(0.062)	0.10	2.07(0.11)	15.695	1.43(0.06)	0.091
80, -80	19.86(4.38)	6.218(2.036)	14.15(0.121)	0.82	1.22(0.17)	10.212	0.81(0.06)	0.079
OMC								
OMC2 (a)	25.90(0.81)	7.391(0.181)	13.95(0.035)	1.45	1.256(0.012)	7.583	4.75(0.04)	0.626
S1 (a)	19.01(0.23)	7.576(0.099)	14.17(0.008)	8.94	0.944(0.016)	9.923	9.41(0.04)	0.948
S2 (a)	23.77(0.87)	6.743(0.152)	14.14(0.029)	4.76	0.637(0.041)	10.699	5.13(0.07)	0.479
S3 (a)	26.33(0.45)	7.135(0.077)	14.17(0.018)	0.41	1.506(0.053)	10.983	7.12(0.05)	0.648
S4 (a)	27.82(0.79)	7.317(0.117)	14.16(0.037)	0.59	1.106(0.071)	10.714	6.56(0.08)	0.612

(a) fits include the (2,2) hyperfine satellites

Table 1b. Fit results to the observed spectra: $\Delta v = 0.3 \text{ km s}^{-1}$; low-density solution

Position	T_k (K)	$\log(\tilde{n}_{\text{H}_2} / \text{cm}^{-3})$	$\log(N_{\text{NH}_3} / \text{cm}^{-2})$	χ^2	Δv_{obs} (km/s)	T_B (K)		η_f
						theor.	obs.	
S106								
200, 40	24.29(2.56)	4.488(0.461)	14.06(0.127)	10.58	1.43(0.08)	6.923	3.04(0.06)	0.439
160, 40	29.14(2.62)	4.180(0.556)	13.84(0.227)	1.31	1.91(0.07)	4.042	2.49(0.08)	0.616
40, 0	30.85(3.21)	4.762(0.801)	14.01(0.207)	1.19	1.58(0.07)	7.577	0.84(0.03)	0.111
0, 40	26.43(1.28)	4.092(0.313)	13.91(0.105)	0.40	1.81(0.05)	4.145	1.38(0.03)	0.333
0, 0 (a)	25.42(1.19)	4.229(0.218)	14.11(0.073)	1.47	1.47(0.02)	6.081	4.71(0.06)	0.776
0, -40	24.26(5.67)	3.813(1.331)	13.78(0.410)	0.29	2.00(0.19)	2.534	1.42(0.12)	0.560
S87								
0,0 21.0 km/s	29.06(3.14)	4.263(0.552)	14.15(0.180)	2.49	1.36(0.07)	6.784	1.10(0.04)	0.162
0,0 23.5 km/s	39.33(2.96)	3.828(0.328)	13.63(0.067)	5.45	2.05(0.07)	2.025	1.28(0.03)	0.632
W48								
-80, 80	23.51(1.42)	4.001(0.4259)	13.69(0.115)	1.09	1.72(0.04)	2.668	2.60(0.05)	0.975
-40, 40	31.04(2.27)	3.651(0.3623)	13.83(0.118)	0.38	2.66(0.09)	2.260	1.40(0.04)	0.619
0, 80	31.32(3.19)	4.197(0.5958)	14.06(0.211)	1.07	2.07(0.11)	5.708	1.43(0.06)	0.251
OMC								
OMC2 (a)	32.66(0.60)	4.490(0.209)	13.60(0.039)	4.85	1.256(0.012)	3.264	4.75(0.04)	(*)
S3 (a)	35.63(0.59)	4.100(0.119)	13.65(0.028)	10.07	1.506(0.053)	2.738	7.12(0.05)	(*)
S4 (a)	40.26(1.11)	4.265(0.223)	13.61(0.041)	5.95	1.106(0.071)	2.873	6.56(0.08)	(*)

(*) the observed brightness temperature is higher than the theoretical one

(a) fits include the (2,2) hyperfine satellites

Table 2a. Derived quantities; $\Delta v = 0.3 \text{ km s}^{-1}$; high-density solution

Position	$\log(\lambda_J/10^{-2} \text{ pc})$	$\log(M_J/M_\odot)$	$\log(M_c/M_\odot)$	$\log(K)$	$\log(v/\text{pc}^{-3})$	$\log(\bar{n}/\text{cm}^{-3})$	$\log([NH_3]/[H_2])$
S106 (r=0.60 kpc, $\Theta_S=1.5^\circ$)							
200, 40	0.296(0.081)	-0.241(0.097)	1.368(0.161)	1.751(0.173)	4.147(0.173)	5.033(0.101)	-8.780(0.085)
200, 80	-0.035(0.666)	-0.664(0.720)	1.496(1.322)	2.145(1.336)	4.541(1.336)	5.003(0.674)	-8.980(0.674)
160, 40	0.193(0.162)	-0.308(0.191)	1.522(0.319)	2.024(0.327)	4.420(0.327)	5.239(0.169)	-8.979(0.172)
160, 0	0.423(0.182)	-0.091(0.194)	0.982(0.361)	1.437(0.365)	3.833(0.365)	4.868(0.184)	-8.845(0.186)
90, 30	1.177(0.743)	0.835(0.762)	-0.762(1.484)	-0.166(1.489)	2.229(1.489)	4.191(0.748)	-8.603(0.797)
40, 0	0.360(0.213)	-0.079(0.252)	0.507(0.419)	1.112(0.430)	3.507(0.430)	4.556(0.221)	-8.875(0.234)
0, 40	0.292(0.087)	-0.245(0.103)	0.895(0.172)	1.541(0.177)	3.936(0.177)	4.818(0.094)	-8.815(0.091)
0, 0 (a)	0.432(0.085)	-0.107(0.092)	1.338(0.168)	1.654(0.170)	4.049(0.170)	5.069(0.087)	-8.573(0.087)
0, -40	1.147(1.720)	0.623(1.758)	-0.522(3.432)	0.061(3.442)	2.457(3.442)	4.206(1.725)	-8.162(1.863)
-40, 0	0.800(0.241)	0.240(0.243)	0.470(0.481)	0.760(0.482)	3.155(0.482)	4.522(0.242)	-8.232(0.243)
S87 (r=1.3 kpc, $\Theta_S=1.5^\circ$)							
0,0 21.0 km/s	0.341(0.127)	-0.156(0.168)	1.521(0.249)	1.774(0.263)	3.163(0.263)	4.134(0.142)	-8.595(0.136)
0,0 23.5 km/s	-0.031(0.110)	-0.493(0.139)	2.537(0.214)	2.855(0.223)	4.243(0.223)	4.877(0.117)	-9.173(0.121)
W48 (r=3.4 kpc, $\Theta_S=3.0^\circ$)							
-80, 40	0.477(0.494)	-0.083(0.511)	2.937(0.985)	2.738(0.989)	2.572(0.989)	3.616(0.497)	-8.706(0.498)
-80, 80	-0.086(0.238)	-0.687(0.251)	4.470(0.474)	4.196(0.477)	4.031(0.477)	4.471(0.240)	-9.318(0.242)
-40, 40	0.149(0.114)	-0.379(0.141)	3.381(0.224)	3.454(0.232)	3.289(0.232)	4.037(0.122)	-8.856(0.120)
0, 40 45km/s	0.881(0.741)	0.512(0.760)	1.731(1.480)	1.625(1.485)	1.460(1.485)	3.099(0.748)	-8.243(0.763)
0, 80	0.382(0.182)	-0.085(0.221)	2.902(0.359)	2.830(0.370)	2.665(0.370)	3.707(0.194)	-8.633(0.192)
80, -80	0.430(1.024)	-0.149(1.070)	2.603(2.039)	2.444(2.054)	2.279(2.054)	3.256(1.036)	-8.744(1.031)
OMC (r=0.5 kpc, $\Theta_S=1.5^\circ$)							
OMC2 (a)	-0.099(0.092)	-0.563(0.102)	2.627(0.181)	2.748(0.184)	5.381(0.184)	5.946(0.093)	-9.588(0.098)
S1 (a)	-0.259(0.050)	-0.857(0.053)	3.082(0.099)	3.124(0.101)	5.757(0.101)	6.027(0.053)	-9.394(0.051)
S2 (a)	0.206(0.078)	-0.295(0.094)	1.804(0.153)	1.727(0.170)	4.360(0.170)	5.192(0.102)	-9.056(0.083)
S3 (a)	0.033(0.039)	-0.424(0.046)	2.393(0.078)	2.579(0.087)	5.212(0.087)	5.915(0.053)	-9.244(0.043)
S4 (a)	-0.046(0.060)	-0.479(0.072)	2.538(0.118)	2.578(0.137)	5.211(0.137)	5.859(0.089)	-9.357(0.071)

Table 2b. Derived quantities; $\Delta v = 0.3 \text{ km s}^{-1}$; low-density solution

Position	$\log(\lambda_J/10^{-2} \text{ pc})$	$\log(M_J/M_\odot)$	$\log(M_c/M_\odot)$	$\log(K)$	$\log(v/\text{pc}^{-3})$	$\log(\bar{n}/\text{cm}^{-3})$	$\log([NH_3]/[H_2])$
S106 (r=0.60 kpc, $\Theta_S=1.5^\circ$)							
200, 40	1.339(0.236)	0.847(0.280)	-0.270(0.462)	-0.066(0.477)	2.329(0.477)	4.303(0.244)	-8.013(0.268)
160, 40	1.532(0.282)	1.120(0.309)	-0.357(0.558)	-0.180(0.565)	2.215(0.565)	4.461(0.286)	-8.118(0.362)
40, 0	1.253(0.404)	0.866(0.430)	-0.892(0.803)	-0.450(0.810)	1.946(0.810)	3.938(0.408)	-8.252(0.454)
0, 40	1.555(0.158)	1.100(0.173)	-0.846(0.315)	-0.517(0.319)	1.879(0.319)	4.106(0.162)	-7.983(0.190)
0, 0 (a)	1.478(0.111)	1.006(0.130)	-0.158(0.219)	-0.086(0.224)	2.310(0.224)	4.443(0.113)	-7.843(0.133)
0, -40	1.676(0.676)	1.184(0.752)	-0.786(1.337)	-0.489(1.357)	1.906(1.357)	4.217(0.688)	-7.955(0.790)
S87 (r=1.3 kpc, $\Theta_S=1.5^\circ$)							
0,0 21.0 km/s	1.490(0.281)	1.076(0.320)	-0.137(0.555)	-0.152(0.566)	1.236(0.566)	3.439(0.288)	-7.849(0.334)
0,0 23.5 km/s	1.773(0.168)	1.491(0.199)	0.315(0.330)	0.051(0.339)	1.439(0.339)	4.057(0.173)	-8.217(0.181)
W48 (r=3.4 kpc, $\Theta_S=3.0^\circ$)							
-80, 80	1.575(0.215)	1.069(0.231)	2.023(0.426)	1.394(0.430)	1.229(0.430)	3.425(0.217)	-8.132(0.244)
-40, 40	1.810(0.185)	1.425(0.212)	1.377(0.365)	0.916(0.372)	0.750(0.372)	3.302(0.190)	-7.878(0.219)
0, 80	1.539(0.302)	1.158(0.334)	1.335(0.598)	0.957(0.607)	0.791(0.607)	3.076(0.309)	-7.922(0.368)

angular diameter θ_s . We then get

$$v = \frac{K}{\Omega_m r^2 \theta_s r}$$

$$= 1.45 \cdot 10^3 \text{ pc}^{-3} \frac{n'_f}{T_1} \frac{\Delta v_{\text{obs}}}{\Delta v} \eta_f \frac{1}{r_{0.5 \text{ kpc}}} \frac{1}{\theta'_s}$$

and from this we can calculate the average density the cloud would have if the clumps are smeared out to form a homogeneous cloud

$$\bar{n}_{H_2} = \frac{1}{2 + 4X_{\text{He}/H_2}} \frac{v M_J}{m_H}$$

$$= 2 \cdot 10^5 \text{ cm}^{-3} (T_1 n'_f)^{1/2} \frac{\eta_f}{r_{0.5 \text{ kpc}} \theta'_s} \frac{\Delta v_{\text{obs}}}{\Delta v}$$

The clumps may have formed by fragmentation out of a cloud with this mean density.

From the fitted column density we may compute the volume density of NH₃ by assuming the line of sight extension of the clump to be again the Jeans length. We then get

$$n_{\text{NH}_3}/n_{\text{H}_2} = \frac{N_{\text{NH}_3}}{\lambda_J n_{\text{H}_2}} = 7.31 \cdot 10^{-24} \frac{N_{\text{NH}_3}}{(T_1 n_1)^{1/2}}.$$

Typical values for the high density solutions are 10^{-8} – 10^{-9} , the values for the low density solutions are one order of magnitude higher. These values are in good agreement with the abundances Graedel et al. (1982) give as results of chemical model calculations with ion-molecule reactions. They give an upper limit of 10^{-8} for $n_{\text{NH}_3}/n_{\text{H}_2}$ where the actual value critically depends on the metal abundance: with high metal abundance the value is 10^{-10} – $5 \cdot 10^{-10}$, with low metal abundance the value is $5 \cdot 10^{-9}$ – 10^{-8} . Again, for the low density solution the agreement is somewhat poorer as the deduced values are somewhat higher than the upper limit of 10^{-8} .

5. Conclusions and discussion

The assumed linewidth of 0.3 km s^{-1} necessary for the selective trapping in the far IR (2,1)→(1,1) line producing the anomalies just equals the sound velocity in the anomalous emitting sources. Larson (1981) gives an empirical relation between the velocity dispersion σ and the cloud diameter L : $\sigma [\text{km s}^{-1}] = 1.1 L [\text{pc}]^{0.38}$, which may be interpreted as resulting from turbulent fragmentation of molecular clouds. He argues, that the fragmentation process should stop, when the velocity dispersion reaches the velocity of sound, as then further turbulence will not lead to dynamical instabilities. The diameter of the smallest fragments reaches $0.3 \cdot 10^{-2} \text{ pc}$ at a temperature of 20 K. This is of the same size as the Jeans length of the anomalous emitting clumps derived in the previous section. Thus it is possible that these clumps are just the end product of turbulent fragmentation.

The turbulence is perhaps initiated by the stellar wind and the associated shock fronts of young OB stars embedded in all anomalous emitting clouds. The reason that otherwise similar warm molecular clouds show no anomalies then may be that these clouds are not yet evolved to the state of many small fragments. Another possibility is that the turbulence arises from the collision of parts of the cloud; this possibility is supported by the fact that many of the anomalous emitting clouds show two velocity components. On the other hand this may be simply a projection effect (as in the case of W48; Morane, 1984).

From the observed linewidth, which must be interpreted as resulting from the average relative velocity of the clumps, and the derived number density we may calculate the collision time for the clumps

$$\tau_{\text{coll}} = \frac{1}{\pi v \lambda_J^2 v_{\text{obs}}} = 2 \cdot 10^5 \text{ yr } (v/10^3 \text{ pc}^{-3})^{-1} (\lambda_J/10^{-2} \text{ pc})^{-2} \cdot [4v_{\text{obs}}/(1.5 \text{ km s}^{-1})]^{-1}.$$

The free fall time of one clump

$$\tau_{\text{ff}} = \left(\frac{3\pi}{32G\rho} \right)^{1/2} = 1.3 \cdot 10^4 \text{ yr } (n_1)^{-1/2}$$

is an order of magnitude smaller, so that the clumps will collapse once they accumulate sufficient mass by collisions. The clumps therefore will collapse after several collision times. The simplified

model calculations of Scalo and Pumphrey (1983) gives a time of $\sim 7\tau_{\text{coll}}$ until the number of fragments has decreased to half the number in the beginning, i.e. after $\sim 7\tau_{\text{coll}}$ the fragments all have undergone collisions leading to coagulation and instability.

The resulting picture for the anomalous emitting NH₃ clouds therefore is the following: a homogeneous cloud of intermediate density (10^4 – 10^5 cm^{-3}) has fragmented, perhaps triggered by the turbulence associated with the stellar wind of a young star or by cloud collisions. The final fragments (clumps) have masses of 0.3 – $1 M_{\odot}$ and a diameter of some 10^{-2} pc ; they represent the low mass end of the mass spectrum, as the velocity dispersion reaches the sound velocity at this clump size. The clumps have thermalized their internal motions, so that their emission shows thermal linewidth. Within 10^5 – 10^6 yr the clumps may coagulate to higher mass fragments which are less stable and at least some of them may collapse to form low mass stars.

Whereas this interpretation may seem somewhat speculative, it has the advantage to be consistent with the simple model assumptions which have been made in order to explain the observed anomalies. Being aware of the fact, that a more sophisticated model probably will change the derived numerical values to some extent, we nevertheless think that the general outline of the given interpretation is correct. Let us review the assumptions made for this interpretation and briefly discuss what the consequences would be, if changes on the assumptions are made, and how these assumptions can be tested by independent observations.

i) The assumed nearly thermal line width of 0.3 km s^{-1} for the individual clumps produces the strongest anomalies. Broader lines would result in less efficient selective trapping and thus produce weaker anomalies. Stated otherwise, somewhat lower densities would be sufficient to reproduce the observed anomalies.

A precise indicator for high densities is the intensity of the non-metastable (2,1) inversion line. Observations of this line therefore provide a good test for the predicted high densities. Another possibility is the direct observation of individual clumps with high angular and frequency resolution. The VLA is just on the limit of sensitivity to perform such observations.

ii) The anomalously emitting clumps are assumed not to be embedded in a normal emitting cloud. In case they were, the observed spectra would be an average of normal and anomalous spectra. The anomalies in the emission of the clumps then had to be stronger than the observed ones. Stronger anomalies require somewhat lower densities. Again the observation of the (2,1) inversion line provides a good test.

iii) The clumps are assumed to have Jeans mass and a diameter given by the Jeans length. Bigger clumps are unstable, resulting in the linewidth broadened by the velocity field arising in the collapse of the clump. The existence of smaller clumps is improbable; even if they exist, they would coagulate rather fast to build up bigger clumps as the collision time $\tau_{\text{coll}} = (\sigma v)^{-1}$ is proportional to the diameter of the clump, when the total mass is held fixed. A direct observational test may be provided by high angular observations with the VLA, which directly allow to determine the clump size.

iv) A possible IR continuum at the (2,1)→(1,1) frequency is neglected in the radiative transfer calculations. Its influence on the anomalies is difficult to predict. A strong IR field will increase the population in the (2,1) levels and therefore lower the optical depth of the IR transition. At a lower optical depth the escape probability is lower also, so that trapping effects have less influence. On the other hand the change of the escape probability with optical depth is greater at lower optical depths, which increases the influence of selective trapping.

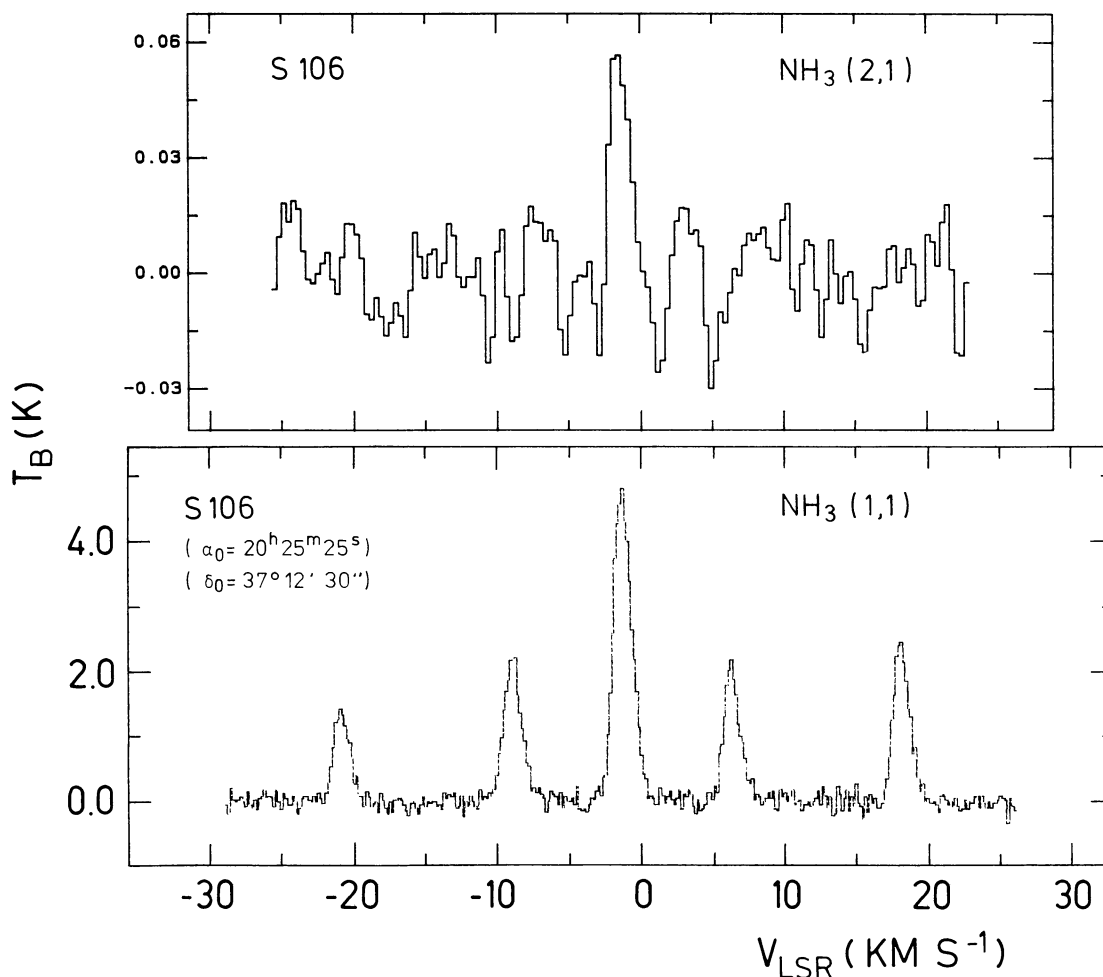


Fig. 7. The observed spectrum of the NH₃ (2,1) inversion transition toward the (1,1) maximum position in S106 and the (1,1) spectrum at the same position

Table 3. Results of NH₃ (2,1) observations

Source	T _B (2,1)	T _B (1,1)	T _B (2,1)/T _B (1,1)	
			theor.	obs.
S106	0.065(0.014)	4.713(0.060)	0.018	0.014(0.003)
OMC S3	0.109(0.017)	7.115(0.049)	0.052	0.015(0.002)
OMC S4	0.186(0.014)	6.558(0.080)	0.068	0.028(0.002)

If the IR radiation would have significant influence on the strength of the anomalies, one should see variations in their strength with the distance from the exciting source. Maps of anomalous sources so far are only available for S106 and W48 and both show no such changes.

Two observational tests therefore are of major importance to examine the presented interpretation of anomalous spectra. One is the measurement of the (2,1) inversion line, which very sensitively tests the predicted high densities. Although for high densities (10^6 cm^{-3}) its intensity is $\sim 10^{-2}$ times that of the (1,1) line, for densities of 10^4 – 10^5 cm^{-3} it is practically undetectable (10^{-4} times the intensity of the (1,1) line).

During recent observations with the Effelsberg 100-m-telescope we detected the (2,1) line in S106 (Fig. 7), OMCS3 and S4. Table 3 summarizes the observational results. In all three sources the linewidth and LSR-velocity are (within the errors) the

same as those of the (1,1) line, which indicates that the (2,1) emission comes from the same gas as the (1,1) line. For S106 the (2,1) intensity is in excellent agreement with the intensity predicted from the radiative transfer model. The predicted density of $10^{6.3} \text{ cm}^{-3}$ is thus safely confirmed. The agreement is somewhat poorer in the case of OMC S3 and S4, as the predicted intensity is a factor of 2–3 to high. The predicted very high densities of greater than 10^7 cm^{-3} must be corrected to densities of several 10^6 cm^{-3} to be in agreement with the observed (2,1) intensity. The reason may be, as was discussed in *i*) and *ii*) above, that either the linewidth of the individual clumps is somewhat larger than 0.3 km s^{-1} or that the clumps are embedded in a normal emitting cloud, so that the observed spectrum is an average of normal and anomalous emission.

The observation and interpretation of several CS transitions in Orion B (Snell et al., 1983), which also shows anomalies in the NH₃ (1,1) line (Stutzki et al., 1984), gives similar high densities and filling factors in agreement with the interpretation given here.

The other observational test is to directly observe the individual clumps with sufficient angular and frequency resolution. VLA observations of DR21 (Matsakis et al., 1981) had not sufficient frequency resolution to resolve velocity structures of 0.3 km s^{-1} ; they nevertheless show several clumps in front of the continuum source. VLA observations of OMCS3 (Harris et al., 1983) had not sufficient angular resolution for resolving individual

clumps of the size predicted here. Thus observations with both sufficient angular and frequency resolution are just at the sensitivity limit of the VLA and therefore will be only possible for the nearest and brightest sources; they will have the great advantage to confirm directly the assumption of individual clumps with small line widths.

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Appendix A: radiative transfer in a homogeneous spherical cloud

The formal solution of the radiative transfer equation giving the intensity I_ν at frequency ν at the place $\mathbf{r}$ going in direction $\mathbf{n}$ (see Fig. A1) is

$$I_\nu(\mathbf{r}, \mathbf{n}) = F_{\nu, S} \exp(-\tau_{\nu 0}) + \int_0^{\tau_{\nu 0}} S_\nu(\tau'_\nu) \exp(-\tau'_\nu) d\tau'_\nu,$$

where $F_{\nu, S}$ is the background radiation at frequency ν along the considered path S . The optical depth is

$$\tau'_\nu = - \int_P^{P'} \chi_\nu(\mathbf{r}) \mathbf{n} \cdot d\mathbf{s}, \quad d\tau_\nu = \chi_\nu(\mathbf{r}) ds$$

with the absorption coefficient χ_ν given by Eq. (3). The source function $S_\nu = \eta_\nu / \chi_\nu$ may be written as the weighted sum of the source functions S_i [Eq. (6)] of the contributing transitions

$$S(\mathbf{r}) = \frac{\sum_j k_j(\mathbf{r}) S_j(\mathbf{r}) \phi_j(\nu - \nu_j)}{\sum_j k_j(\mathbf{r}) \phi_j(\nu - \nu_j)}.$$

The mean intensity at frequency ν is obtained by averaging over the angular dependence

$$J_\nu(\mathbf{r}) = \frac{1}{4\pi} \int I_\nu(\mathbf{r}, \mathbf{n}) d^2\mathbf{n},$$

the line intensity in transition t , entering the rate equations (7), is calculated by integrating over the line profile

$$J_t(\mathbf{r}) = \int \frac{\phi_t(\nu - \nu_t)}{\Delta\nu_t} J_\nu(\mathbf{r}) d\nu.$$

In the case of a homogeneous medium τ_ν is proportional to the path length; for a spherical cloud the length s for a path crossing the radial direction at r at an angle of ϑ (see Fig. A2) is

$$s = r \cos \vartheta + \sqrt{R^2 - r^2 \sin^2 \vartheta}$$

and

$$I_\nu(r, \vartheta) = F_\nu \exp[-\chi_\nu(r \cos \vartheta + \sqrt{R^2 - r^2 \sin^2 \vartheta})] + S_\nu \{1 - \exp[-\chi_\nu(r \cos \vartheta + \sqrt{R^2 - r^2 \sin^2 \vartheta})]\}, \quad (\text{A1})$$

where we have assumed isotropic background radiation. The mean intensity then is

$$J_\nu(r) = F_\nu \beta_\nu(r) + S_\nu(1 - \beta_\nu(r)),$$

where

$$\beta_\nu(r) = \frac{1}{2} \int_{-1}^{+1} \exp[-\chi_\nu(r\mu + \sqrt{R^2 - r^2 + r^2\mu^2})] d\mu, \quad \mu = \cos \vartheta$$

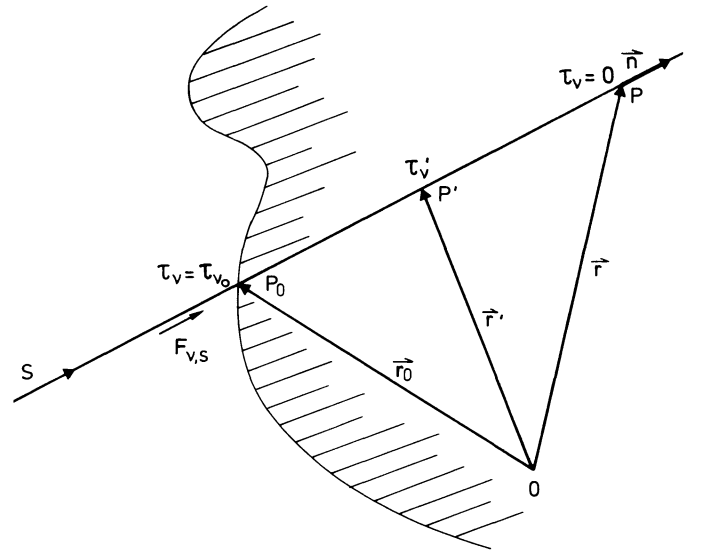


Fig. A1. Radiative transfer: general geometry

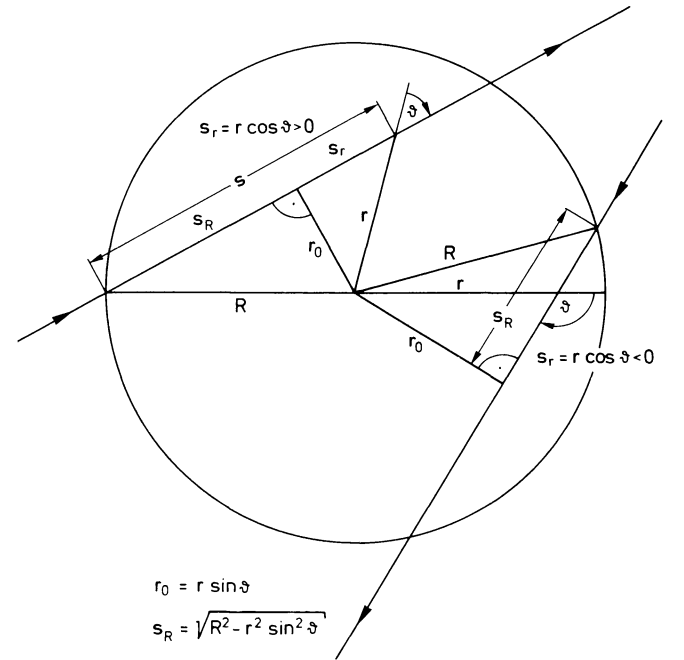


Fig. A2. Radiative transfer in a spherical cloud

is the escape probability for a photon with frequency ν originating at r . The angular integration may be carried out analytically (Stutzki, 1984). The limiting cases give

$$\beta_\nu(r=0) = \exp(-\chi_\nu R)$$

and

$$\beta_\nu(r=R) = \frac{1}{2} \left(1 + \frac{1 - \exp(-2\chi_\nu R)}{2\chi_\nu R} \right),$$

which leads to the limit 1/2 when the cloud becomes optically thick, i.e. the photon can escape only in the halfspace away from the cloud.

Assuming that the background intensity is constant over the line profile, the frequency integration necessary to obtain the line intensity J_t gives for the background contribution

$$\int \frac{\phi_t(v-v_t)}{\Delta v_t} F_v \beta_v(r) dv = F_{v_t} \beta_t(r),$$

where

$$\beta_t(r) = \int \frac{\phi_t(v-v_t)}{\Delta v_t} \beta_v(r) dv \quad (\text{A2})$$

is the escape probability in line t . The frequency integration on the part containing the source function is more complicated:

$$\int \frac{\phi_t(v-v_t)}{\Delta v_t} S_v(1-\beta_v(r)) dv = \sum_j (\alpha_{tj} - \varepsilon_{tj}) S_j, \quad (\text{A3})$$

where

$$\alpha_{tj} = \int \frac{\phi_t(v-v_t)}{\Delta v_t} \frac{k_j \phi_j(v-v_j)}{\sum_j k_j \phi_j(v-v_j)} dv,$$

$$\varepsilon_{tj} = \int \frac{\phi_t(v-v_t)}{\Delta v_t} \frac{k_j \phi_j(v-v_j)}{\sum_j k_j \phi_j(v-v_j)} \beta_v(r) dv$$

are the photon exchange- and exchange-escape-probabilities between transition t and j satisfying $\sum_j \alpha_{tj} = 1$ and $\sum_j \varepsilon_{tj} = \beta_t$. For the simplified model used in this paper, we assume that either two transitions do not overlap or they have the same frequency, i.e. they belong to the same transition group G . Only transitions in the same line group as t then contribute to the j -sum in (A3) and all these transitions have the same profile ϕ so that their contributions cancel in the nominator and denominator in (A3).

The final result for the frequency integrated source function then is

$$\int \frac{\phi_t(v-v_t)}{\Delta v_t} S_v(1-\beta_v(r)) dv = S_G(1-\beta_t(r)),$$

where S_G is the source function of transition group G [see Eq. (5)].

The observable intensity, expressed as main beam brightness temperature, is calculated from the intensity at the cloud surface $I_v(R, \vartheta)$ [Eq. (A1)] by integrating over the beam profile $\psi(\theta)$

$$T_{B;t} = \frac{c^2}{2k\nu_t^2} \left\{ \frac{1}{\Omega_m} \int_{\Omega_m} I_{v_t}(\theta) \psi(\theta) d\Omega - F_{v_t} \right\}, \quad (\text{A4})$$

where $\Omega_m = \int \psi(\theta) d\theta$ is the beam solid angle. If the source is much smaller than the beam, we may put $\psi(\theta) \sim \psi(0) = 1$ and restrict the integration to the source extension. As the distance r to the source is much larger than the source radius R , we have $d\Omega = (R/r)^2 \sin \vartheta \cos \vartheta d\vartheta d\varphi$ and get

$$T_{B;t} = \frac{c^2}{2k\nu_t^2} \frac{\Omega_s}{\Omega_m} 2 \left\{ \int_0^{\pi/2} I_{v_t}(R, \vartheta) \sin \vartheta \cos \vartheta d\vartheta - F_{v_t} \right\}.$$

Inserting (A1) and carrying out the angular integration we obtain Eq. (8). The usual detection equation [Eq. (9)] is obtained, if the source extension is much larger than the beam, so that the source intensity in the beam may be taken out of the integral as a constant factor.

References

- Batrla, W., Wilson, T.L., Bastien, P., Ruf, K.: 1983, *Astron. Astrophys.* **128**, 279
- Green, S.: 1981, *NASA Tech. Memo.* **83869**
- Graedel, T.E., Langer, W.D., Frerking, M.A.: 1982, *Astrophys. J. Suppl.* **48**, 321
- Guilloteau, S., Baudry, A.: 1981, *Astron. Astrophys.* **97**, 213
- Harris, A., Townes, C.H., Matsakis, D.N., Palmer, P.: 1983, *Astrophys. J. Letters* **265**, L 63
- Lang, R.: 1980, *Astrophysical Formulae*, Springer, Berlin, Heidelberg, New York
- Larson, R.B.: 1981, *Monthly Notices Roy. Astron. Soc.* **194**, 809
- Matsakis, D.N., Brandshaft, D., Chui, M.F., Cheung, A.C., Yngvesson, K.S., Cardiasmenos, A.G., Shanley, J.F., Ho, P.T.P.: 1977, *Astrophys. J. Letters* **214**, L 67
- Matsakis, D.N., Hjalmarsen, A., Palmer, P., Cheung, A.C., Townes, C.H.: 1981, *Astrophys. J. Letters* **250**, L 85
- Morane, A.: 1983, Thèse de 3^e cycle, Grenoble
- Scalo, J.M., Pumphrey, W.A.: 1982, *Astrophys. J. Letters* **258**, L 29
- Snell, R.L., Mundy, L.G., Goldsmith, P.F., Evans, N.J., Erickson, N.R.: 1984, *Astrophys. J.* **276**, 625
- Stutzki, J., Ungerechts, H., Winnewisser, G.: 1982, *Astron. Astrophys.* **111**, 201
- Stutzki, J., Jackson, J.M., Olberg, M., Barrett, A.H., Winnewisser, G.: 1984, *Astron. Astrophys.* **139**, 258
- Stutzki, J., Winnewisser, G.: 1984, *Astron. Astrophys.* (in press)
- Stutzki, J., Winnewisser, G.: 1985, *Astron. Astrophys.* (submitted)
- Townes, C.H., Schawlow, A.L.: 1955, *Microwave Spectroscopy*, Dover Publ., New York